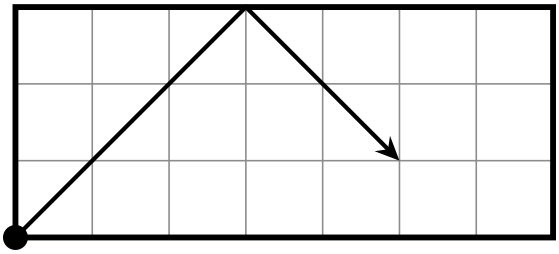
The ball starts at the dot in the bottom left corner of a table and rolls up and to the right, crossing each square from corner to corner. It bounces off every wall it hits and stops when it reaches a corner of the table. Each time the ball hits a wall counts as one bounce; reaching the corner at the end does not count. A 7 by 3 table is 7 squares wide and 3 squares high.



7 by 3

Before one of you traces a path, the others predict where the ball will stop. Then they check the tracing.

Problem 6: Find a table for each of these, or explain why there is none.

- The ball stops at the top left corner after 13 bounces.
- The ball stops at the top right corner after 8 bounces.
- The ball stops at the bottom right corner after 11 bounces.
- The ball stops at the bottom right corner after 10 bounces.

Problem 7: Is there a table where the ball stops at the corner it started from? Find one, or explain why there is none.

Problem 8: Write a rule that tells where the ball stops and how many times it bounces on any table, without tracing. Explain why your rule works.